REAL-TIME BIOMECHANICAL EXERCISE CORRECTION USING COMPUTER VISION AND MULTILINGUAL VOICE FEEDBACK FOR ATHLETIC TRAINING

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Abstract

A critical gap exists in athletic training environments: most athletes have no access to objective, joint-level analysis of their exercise technique in real time, leaving them dependent on coach observation that is inherently subjective and limited in capacity. Repeated training with poor biomechanical alignment causes compensatory movement patterns including knee valgus, excessive trunk lean, and hip sagging that elevate injury risk while diminishing performance gains. This work introduces NeuroMotion, a camera-based system that employs two custom-trained YOLOv8 deep learning models to simultaneously classify exercise type and estimate 33 skeletal keypoints. Vector-based trigonometry is applied to compute joint angles, which are assessed against exercise-specific biomechanical thresholds for squats, push-ups, and jumps. Immediate correction is provided through a color-coded skeleton overlay and multilingual spoken guidance in English, Tamil, and Sinhala. Model training used a dataset of over 500 video clips from more than 20 volunteers, divided 70/15/15 for training, validation, and testing. Evaluation showed 92.1% posture detection accuracy, 94.8% repetition counting accuracy, and frame processing latency below 100ms. The system requires only a standard webcam, making biomechanical coaching accessible without wearable sensors or motion capture equipment. Future work should target exercise library expansion, 3D depth integration, and scalable cloud deployment for institutional use.

Keywords: pose estimation, YOLOv8, biomechanical analysis, real-time posture correction, multilingual voice feedback

Dedication

I dedicate this dissertation to my beloved family, whose unwavering support, encouragement, and sacrifices have been the foundation of my academic journey. To my parents, who instilled in me the value of education and hard work, and to my siblings, who have always believed in my potential, this achievement belongs to you as much as it does to me.

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List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CPU	Central Processing Unit
CVAT	Computer Vision Annotation Tool
EMA	Exponential Moving Average
FC	Fully Connected
FN	False Negative
FP	False Positive
FPN	Feature Pyramid Network
FPS	Frames Per Second
GPU	Graphics Processing Unit
IMU	Inertial Measurement Unit
LiDAR	Light Detection and Ranging
ReLU	Rectified Linear Unit
SaaS	Software as a Service
SLIIT	Sri Lanka Institute of Information Technology
TTS	Text-to-Speech
YOLOv8	You Only Look Once version 8

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1. INTRODUCTION

1.1 Background and Literature Survey

Sports training environments, ranging from university athletics programs to commercial fitness centers, rely heavily on coach observation for technique monitoring. In these settings, trainers visually assess athlete performance during resistance training, plyometric drills, and conditioning work. While experienced coaches can identify many form errors, the nature of human observation introduces inherent limitations that can compromise training quality and athlete safety. The challenge of maintaining proper exercise form becomes particularly acute when coaches must monitor multiple athletes simultaneously or when movements occur too rapidly for accurate visual assessment. This limitation is especially pronounced in settings where a single coach oversees multiple athletes during group training sessions, a common scenario in most sports academies and commercial gym environments.

The consequences of training with incorrect posture extend beyond immediate discomfort. Habitual movement patterns such as knee valgus, where knees cave inward during squats, excessive forward lean that places undue stress on the lumbar spine, and hip sagging during planks or push-ups can develop over time. These compensatory patterns not only reduce training effectiveness but also significantly increase the risk of both acute injuries and chronic overuse conditions. Training with improper form is widely recognized in sports medicine to increase injury risk and reduce training effectiveness. Studies have shown that athletes who train with improper form for extended periods develop muscle imbalances and movement compensations that can take months or years to correct, making early detection and correction of form errors critically important.

Traditional approaches to exercise form correction have relied primarily on the expertise of trained coaches and physical therapists. These professionals use their trained eyes to identify posture deviations and provide verbal or tactile cues to guide athletes toward correct form. However, this approach has several fundamental limitations that constrain its effectiveness in modern training environments. First, even

highly skilled coaches cannot observe multiple athletes simultaneously with equal attention to detail, creating blind spots in group training scenarios. Second, fast movements, such as the bottom phase of a squat or the transition in a push-up, occur too quickly for the human eye to catch subtle posture errors in real-time. Third, visual assessment is inherently subjective, and different observers may reach different conclusions about the same movement, leading to inconsistent feedback for athletes.

The subjectivity inherent in visual assessment presents a particular challenge for athletes seeking consistent, objective feedback on their technique. Two different coaches observing the same athlete might offer conflicting advice on form, creating confusion and potentially hindering progress. Furthermore, the quality of coaching feedback depends heavily on the coach's experience level, training background, and current attention state. A fatigued coach at the end of a long training day may miss errors that would be immediately apparent at the start of the session. This variability in human observation quality motivates the development of automated, objective assessment systems that can provide consistent feedback regardless of time, attention state, or observer experience.

Recent advances in technology have introduced various solutions attempting to address these limitations. Wearable sensor systems, including accelerometer-based fitness trackers and inertial measurement units (IMUs), can capture movement data with high precision. Chen et al. developed a multi-sensor system for squat performance assessment that achieved 89% accuracy in detecting major form errors [1]. Their research demonstrated that IMU-based systems could effectively detect gross movement patterns and timing errors in squat performance. However, these systems typically require multiple sensors attached to different body parts and cannot reconstruct skeletal geometry from inertial data alone, limiting their ability to detect joint-specific posture errors such as knee valgus alignment or proper hip positioning during movement.

The fundamental limitation of wearable systems stems from the nature of inertial sensor data. Accelerometers measure linear acceleration, and gyroscopes measure angular velocity, but neither directly provides positional information. Reconstructing

body segment positions from IMU data requires integration over time, which accumulates drift errors that degrade accuracy during extended movements. More critically, determining whether the knee is properly aligned over the ankle requires knowing the relative positions of these joints, information that IMU sensors cannot directly provide. Some advanced wearable systems attempt to address this through sensor fusion algorithms, but the accuracy remains inferior to vision-based approaches for joint-level posture assessment.

Motion capture laboratories using marker-based systems like Vicon or OptiTrack provide highly accurate kinematic data and represent the gold standard for biomechanical analysis. Dressler et al. demonstrated that such systems can achieve sub-millimeter accuracy in tracking joint positions during athletic movements [3]. These systems use multiple synchronized cameras to track reflective markers placed on the athlete's body, enabling precise three-dimensional reconstruction of body segment positions and orientations. The resulting kinematic data provides detailed information about joint angles, velocities, and accelerations that is invaluable for both research and clinical applications. However, motion capture laboratories require controlled environments with specialized equipment costing tens of thousands of dollars, extensive technical expertise to operate, and significant setup time for each session.

The practical limitations of motion capture laboratories effectively restrict access to professional and elite athletes while excluding the vast majority of recreational athletes and fitness enthusiasts. Setting up a motion capture session requires attaching reflective markers to specific anatomical landmarks on the athlete's body, a process that can take 30 minutes or longer for a full-body capture. The controlled environment requirement means that data cannot be collected during actual training sessions in typical gym settings. These constraints make motion capture impractical for routine training feedback, limiting its use primarily to research laboratories and specialized sports science facilities serving elite athletes.

Consumer fitness applications have proliferated in recent years, offering features such as repetition counting, workout tracking, and guided exercise programs. These

applications leverage the ubiquity of smartphones to provide fitness guidance at minimal cost, reaching millions of users worldwide. While these applications have made fitness more accessible and engaging for many users, they typically focus on quantity rather than quality of movement. Liao et al. proposed a deep learning framework for assessing rehabilitation exercise quality, demonstrating the feasibility of automated quality evaluation but limiting application to rehabilitation contexts rather than athletic training [4]. An athlete might complete ten squats with severely compromised form, and the application would register this as a successful set without providing any corrective feedback on the biomechanical quality of the movements performed.

The absence of quality assessment in consumer fitness applications represents a significant missed opportunity for injury prevention and training optimization. Many common injuries in fitness training result from cumulative microtrauma caused by repeated movements with suboptimal form. Lower back pain, knee injuries, and shoulder impingements often develop gradually from training with postures that place excessive stress on vulnerable structures. Applications that count repetitions without assessing form quality may inadvertently encourage athletes to prioritize quantity over quality, potentially increasing injury risk rather than reducing it. This gap between available technology and effective biomechanical feedback motivates the development of systems that combine accessibility with meaningful posture assessment.

Figure 1.1 illustrates the fundamental limitations inherent in existing coaching and training systems. The diagram categorizes these limitations across four dimensions: accuracy, accessibility, real-time feedback capability, and cost-effectiveness. Traditional coach observation, while accessible in team settings, lacks the precision to detect subtle biomechanical errors and suffers from subjectivity and attention limitations. Wearable systems improve accuracy for certain metrics but sacrifice accessibility due to sensor requirements and cannot provide direct joint alignment assessment. Motion capture laboratories excel in accuracy but fail on accessibility and cost dimensions, limiting their use to specialized facilities. Consumer applications address accessibility but provide no meaningful feedback on movement quality, leaving athletes without guidance on proper form.

Figure 1.1: Overview of Limitations in Existing Coaching Systems Source: Author

Table 1.1 presents a comprehensive comparison of existing exercise analysis systems against the proposed NeuroMotion approach. The comparison evaluates systems across seven key dimensions: exercise-specific corrective feedback, camera-only operation, joint angle analysis capability, exercise-specific thresholds, multilingual voice feedback, performance dashboard availability, and affordability for individual athletes. This comparison reveals a clear gap in existing solutions that the present research aims to address. No existing system combines all seven capabilities, with most systems offering only a subset of the features necessary for comprehensive real-time posture correction.

Table 1.1: Comparison of existing exercise analysis systems

Feature	**Wearable**	**MediaPipe**	**Motion Capture**	**NeuroMotion**
Exercise-specific corrective feedback	No	No	No	Yes
Camera only (no wearables)	No	Yes	No	Yes
Joint angle analysis	Partial	No	Yes	Yes
Exercise-specific thresholds	No	No	Yes	Yes
Multilingual voice feedback	No	No	No	Yes
Performance dashboard	No	Partial	Yes	Yes
Affordable for individuals	No	Yes	No	Yes

1.2 Research Gap

The literature review reveals a significant gap in the landscape of exercise monitoring and correction technologies. While existing systems address isolated components of exercise monitoring, such as detecting movement patterns through wearables, extracting keypoints through pose estimation, or classifying exercise types through convolutional neural networks, no integrated solution currently exists that combines all necessary capabilities for effective real-time posture correction. This

research identifies four critical gaps that existing technologies fail to address comprehensively, each representing a fundamental limitation that prevents current systems from providing complete athletic coaching support.

The first gap concerns the combination of camera-only operation with joint-level posture feedback. Wearable sensor systems, despite their accuracy in tracking gross movement patterns, cannot reconstruct skeletal geometry and therefore cannot detect joint-specific errors such as knee valgus or excessive forward lean. The relationship between hip, knee, and ankle positions that determines knee valgus alignment cannot be directly measured by accelerometers or gyroscopes. Motion capture laboratories can provide this information but require expensive equipment and controlled environments that limit accessibility. Computer vision approaches using pose estimation algorithms like OpenPose [5] and MediaPipe [6] can extract skeletal information from standard camera feeds but typically stop at keypoint detection without providing exercise-specific biomechanical evaluation. These systems tell us where the body parts are but not whether they are positioned correctly for the exercise being performed.

The second gap relates to exercise-specific biomechanical thresholds. Generic pose estimation provides information about body configuration but does not inherently indicate whether a particular configuration represents correct or incorrect form for a specific exercise. A knee angle of 90 degrees might be correct at the bottom of a squat but incorrect during a jump landing. The trunk angle that indicates proper form for a deadlift differs from the proper trunk angle for a squat. The integration of exercise classification with exercise-specific biomechanical rules represents an underexplored area in current research. Pestana et al. acknowledged that their deep learning system for exercise classification could not distinguish between properly and improperly executed movements [10], highlighting this critical limitation.

Without exercise-specific thresholds, pose estimation systems provide data without actionable interpretation. Knowing that a knee angle is 85 degrees is meaningless without context about what the expected angle should be for the current exercise and phase of movement. The present research addresses this gap by

integrating exercise classification with comprehensive biomechanical threshold databases for each supported exercise. This integration enables the system to not only detect body configuration but also evaluate whether that configuration meets established criteria for correct form, transforming raw pose data into actionable coaching guidance.

The third gap involves the delivery of corrective feedback in real-time. Most existing research systems output their analysis to displays or logs for post-session review, providing information that can inform future training sessions but cannot guide immediate corrections. However, athletes engaged in exercise cannot continuously monitor a screen, limiting the utility of visual feedback during active training. The moment when a posture error occurs is precisely when feedback is most valuable for motor learning and injury prevention. Feedback delivered after the session ends loses the temporal context that makes it actionable, requiring athletes to remember and mentally replay their movements to apply the guidance.

Voice-based feedback represents a natural solution to this timing challenge, yet existing systems have not integrated multilingual speech synthesis capabilities with biomechanical analysis. The human coaching model relies heavily on verbal cues precisely because athletes can process auditory information while executing movements. When a coach says "keep your chest up" during a squat, the athlete can immediately adjust without stopping to look at a screen. This verbal feedback modality has been refined through decades of coaching practice but remains absent from most technological solutions. Furthermore, the multilingual dimension is particularly significant in diverse regions like Sri Lanka, where athletes may prefer coaching in Tamil or Sinhala rather than English.

The fourth gap pertains to accessibility and affordability. The systems that provide the most accurate biomechanical analysis, namely motion capture laboratories, are prohibitively expensive for most athletes and training facilities. A complete Vicon system with necessary cameras, software, and support equipment can exceed one hundred thousand dollars, plus ongoing maintenance and calibration costs. The systems that are affordable, namely consumer fitness applications, provide no

meaningful posture feedback. This creates a situation where high-quality biomechanical coaching is available only to elite athletes with institutional support, while recreational athletes must rely on their own judgment or occasional professional consultations. There exists a need for a solution that achieves professional-grade analysis while remaining accessible to athletes at all competitive levels through the use of standard cameras and commodity computing hardware.

1.3 Research Problem

Athletes in training environments, from sports academies to commercial fitness centers, lack access to real-time, joint-level feedback on exercise form. This deficiency leads to training with incorrect posture, which in turn causes habitual movement patterns such as knee valgus, excessive forward lean, and hip sagging. These patterns increase the risk of both acute injuries during training sessions and chronic overuse conditions over time, while simultaneously reducing training effectiveness and athletic performance development. The prevalence of these movement errors across all levels of athletic training indicates a systemic problem requiring a technological solution that scales beyond individual coaching capacity.

The core research problem can be articulated as follows: How can computer vision technology be leveraged to provide real-time, biomechanically accurate exercise correction feedback that is accessible to athletes at all competitive levels without requiring specialized hardware such as wearable sensors or motion capture equipment? This problem encompasses several technical challenges, including the need for accurate pose estimation under varied conditions, the computational efficiency required for real-time processing, the biomechanical knowledge necessary for meaningful posture evaluation, and the user interface design considerations for delivering actionable feedback during active exercise.

The problem statement addresses multiple stakeholder needs simultaneously. For individual athletes, it promises access to coaching-quality feedback without the cost of personal training sessions or expensive equipment. For coaches, it offers a tool that extends their observational capabilities, allowing them to provide more consistent

feedback across more athletes. For sports academies and training facilities, it provides a scalable solution that can enhance the quality of training across large athlete populations. For rehabilitation professionals, it offers objective movement assessment that can track patient progress over time.

The present research addresses this problem through the development of NeuroMotion, a camera-based exercise correction system that integrates exercise classification, 33-keypoint pose estimation, joint angle calculation using vector trigonometry, exercise-specific biomechanical threshold evaluation, visual feedback through skeleton overlay, and voice-based corrective feedback in multiple languages. The system is designed to operate on standard computing hardware with a standard webcam, making professional-grade biomechanical coaching accessible to athletes who previously could not afford or access such technology. By addressing all four identified gaps in existing technology, the NeuroMotion system represents a comprehensive solution to the stated research problem.

1.4 Research Objectives

The primary objective of this research is to design and develop a real-time, camera-based biomechanical exercise correction system using custom-trained YOLOv8 deep learning models that delivers immediate corrective feedback through visual overlays and multilingual voice synthesis. This main objective encompasses the complete pipeline from video input through feedback delivery, requiring integration of multiple technical components into a cohesive, functional system. The primary objective specifies the key technical approach (YOLOv8 models), the operating mode (real-time, camera-based), the feedback mechanisms (visual overlays and voice synthesis), and the distinguishing feature (multilingual support).

This main objective is supported by three specific objectives that together constitute the complete research contribution. Each specific objective addresses a distinct technical challenge and produces a deliverable component of the overall system. The decomposition into specific objectives facilitates project management,

enables parallel development where appropriate, and provides clear milestones for assessing research progress.

The first specific objective (SO1) is to build a data pipeline for live camera and uploaded video input processing. This objective encompasses the development of infrastructure for capturing real-time video streams from connected cameras, processing uploaded video files for post-session analysis, and managing the flow of visual data through the processing pipeline. The data pipeline must handle varying input resolutions, frame rates, and lighting conditions while maintaining consistent output for downstream processing modules. Technical considerations include video decoding efficiency, frame buffering strategies, and synchronization between input capture and processing threads.

The data pipeline component serves as the foundation for all subsequent processing. Without reliable, efficient video input handling, downstream analysis cannot function effectively. The pipeline must gracefully handle common input challenges such as variable lighting, motion blur, partial occlusion, and frame drops. It must also provide appropriate abstractions that allow the analysis modules to operate independently of the specific input source, enabling the same processing logic to handle both live camera feeds and recorded video files. The deliverable for this objective is a reusable input handling module that can be integrated with various analysis backends.

The second specific objective (SO2) is to develop a computer vision module for 33-keypoint skeleton visualization, joint angle calculation, and guideline line display. This objective involves the implementation of pose estimation using custom-trained YOLOv8 models, the mathematical framework for computing joint angles from detected keypoints using vector trigonometry, the establishment of exercise-specific biomechanical thresholds for posture evaluation, and the visual feedback system including color-coded skeleton overlays and three reference guideline lines. This objective represents the core analytical capability of the system.

The computer vision module transforms raw video frames into actionable biomechanical insights. The pose estimation component must achieve sufficient

accuracy for meaningful joint angle measurement while maintaining real-time processing speed. The joint angle calculation must correctly handle the mathematical challenges of angle computation from 2D projections of 3D movements, including proper handling of angle ambiguity and normalization. The posture evaluation logic must accurately map computed joint angles to posture quality assessments using biomechanically sound thresholds. The visual feedback must provide clear, intuitive indication of posture status without distracting from the exercise performance.

The third specific objective (SO3) is to integrate AI based performance prediction and a feedback dashboard. This objective extends the core posture correction functionality with additional features that enhance the training experience. The performance prediction component analyzes posture patterns over time to predict potential issues before they manifest as injuries, providing proactive guidance rather than reactive correction. The dashboard provides athletes and coaches with session summaries and trend analysis, enabling tracking of progress over multiple sessions.

These extended features differentiate the system from simple real-time feedback tools by providing longitudinal analysis and interactive guidance. The performance prediction capability transforms accumulated movement data into forward-looking insights, helping athletes and coaches identify developing issues before they cause problems. The dashboard enables goal-setting and progress tracking that supports long-term training planning. Together, these features extend the system beyond real-time correction by providing longitudinal performance insights for ongoing training improvement.

2. LITERATURE REVIEW

2.1 Wearable-Based Motion Analysis

Inertial measurement unit sensors have been extensively deployed for human motion capture in sports and rehabilitation contexts. These sensors, which typically combine accelerometers, gyroscopes, and sometimes magnetometers, can be attached to various body segments to track orientation and movement in three-dimensional space. The proliferation of wearable technology in consumer devices has accelerated research in this area, with numerous studies demonstrating the potential for automated movement assessment using IMU data. The miniaturization of IMU sensors and their integration into consumer devices has made motion sensing accessible beyond specialized laboratory settings.

O'Reilly et al. developed a multi-sensor system specifically designed for squat performance assessment, which achieved 89% accuracy in detecting major form errors [2]. Their system employed five body-worn IMU sensors positioned at key anatomical locations including the thighs, shanks, and torso. The sensors captured acceleration and angular velocity data during squat performance, which was then processed through a machine learning classifier trained to identify common form errors including excessive forward lean, knee valgus, and improper depth. The classification approach used time-domain and frequency-domain features extracted from the sensor data streams.

While the Chen et al. system demonstrated promising results for detecting gross movement patterns, the authors acknowledged a significant limitation: the inability to assess knee valgus alignment due to the fundamental constraint that IMU sensors cannot reconstruct skeletal geometry from inertial data alone. Knee valgus, the inward collapse of the knee toward the midline of the body, is a critical error pattern associated with anterior cruciate ligament injury risk. Detecting this error requires knowing the relative positions of the hip, knee, and ankle in the frontal plane, information that cannot be directly extracted from accelerometer and gyroscope data. The system could

detect that the thigh was rotating but could not determine the spatial relationship between knee and ankle positions that defines valgus alignment.

O'Reilly et al. extended this research by investigating wearable systems for injury prevention in athletes across multiple exercise modalities [2]. Their study focused on detecting hazardous movement patterns associated with increased injury risk, including excessive trunk flexion during squats, improper landing mechanics during jumping exercises, and compensatory movements during single-leg exercises. The system achieved high sensitivity for detecting these patterns through machine learning classification of IMU data, with the specific sensitivity varying by error type and sensor configuration. The authors used a combination of rule-based detection for specific patterns and machine learning classification for more complex movement assessments.

The O'Reilly et al. study highlighted that while IMU-based systems excel at capturing movement timing and general patterns, they struggle with exercises requiring precise joint alignment evaluation. The relationship between sensor orientation and joint angle depends on sensor placement, soft tissue movement during exercise, and individual anatomical variations. These factors introduce uncertainty in the derived joint angles that can exceed the tolerances needed for meaningful posture assessment. Furthermore, detecting errors like knee valgus, which involves mediolateral displacement of the knee relative to the hip-ankle line, requires positional information that IMU sensors inherently cannot provide.

The fundamental limitation of wearable approaches lies in their inability to directly measure joint angles and skeletal alignment. An IMU sensor attached to the thigh reports the orientation of the thigh segment in space, but determining whether the knee joint exhibits proper alignment requires relating the thigh orientation to the shank orientation with sufficient accuracy. Sensor placement errors, soft tissue movement during exercise, and drift in IMU orientation estimates all contribute to uncertainties in the derived joint angles. Even small errors in sensor orientation can propagate into large errors in computed joint angles, particularly for movements involving significant rotation.

Several technical approaches have been proposed to address these limitations. Sensor fusion algorithms combine data from multiple IMUs to improve orientation estimation accuracy. Machine learning models learn mappings from IMU data to joint angles using training data from motion capture systems. Magnetometers provide absolute heading reference to reduce drift. However, each of these approaches introduces additional complexity and cost while still falling short of the accuracy achievable through vision-based methods for joint-level assessment. The fundamental constraint remains that inertial sensors measure movement relative to their own reference frame, not absolute positions in space.

Despite these limitations, wearable sensors remain valuable for certain applications. Their portability, battery operation, and independence from environmental constraints make them suitable for field-based assessments and continuous monitoring during training in outdoor or non-standard environments. Several commercial systems have successfully integrated IMU sensors for applications such as running gait analysis, where the focus is on temporal patterns rather than precise joint angles, and general activity tracking, where exact posture is less critical than activity type classification. However, for applications requiring precise joint-level posture assessment, such as the real-time biomechanical exercise correction targeted by the present research, purely IMU-based approaches face fundamental constraints that limit their effectiveness.

Table 2.1 summarizes key wearable-based motion analysis research, highlighting the methodologies, accuracy results, and limitations of each approach. The table demonstrates the consistent pattern of high accuracy for certain metrics combined with fundamental limitations for joint-level assessment. This pattern supports the conclusion that while wearable sensors have an important role in motion analysis, they cannot provide the comprehensive joint-level posture feedback required for effective exercise correction without additional sensing modalities.

Table 2.1: Summary of wearable based motion analysis research

Study	**Method**	**Accuracy**	**Limitation**
Chen et al. [1]	5 IMU sensors	89%	No knee valgus detection

Study	Method	Accuracy	Limitation
O'Reilly et al. [2]	Multiple IMUs	High sensitivity	Joint alignment uncertainty
Ignatov [9]	CNN + accelerometer	>95%	No quality assessment

2.2 Computer Vision and Pose Estimation

Markerless pose estimation has advanced significantly with the introduction of deep learning architectures capable of detecting human body keypoints directly from monocular video. Unlike wearable systems that require sensors attached to the body, computer vision approaches analyze the visual appearance of the human figure to infer body configuration. This paradigm shift has enabled pose estimation systems that operate with nothing more than a standard camera, dramatically reducing barriers to deployment and improving user acceptance. The elimination of worn sensors improves convenience and eliminates the potential for sensors to affect movement patterns.

The development of effective markerless pose estimation has been enabled by several key advances in deep learning. Convolutional Neural Networks (CNNs) have proven highly effective at learning visual representations of human bodies and body parts from large datasets of annotated images. The availability of large-scale annotated datasets such as COCO [12] has provided the training data necessary for learning robust pose estimation models. Advances in GPU computing have enabled training and inference of computationally intensive deep learning models at scales previously infeasible. Together, these advances have transformed pose estimation from a research curiosity into a practical technology.

OpenPose, introduced by Cao et al., pioneered real-time multi-person keypoint detection using an approach called Part Affinity Fields [5]. The system detects 18 body keypoints representing major joints and facial features, then uses learned associations between keypoints to construct coherent skeleton structures even when multiple people are visible in the frame. The Part Affinity Fields approach learns to predict 2D vector fields that encode the position and orientation of body part pairs, enabling the system to associate detected keypoints with the correct body even under occlusion and close

proximity of multiple people. The method achieves real-time performance on modern GPUs and has become a foundational reference in the field.

The OpenPose architecture consists of two branches of a multi-stage convolutional network. The first branch predicts confidence maps for each keypoint, indicating the likelihood of each pixel containing that keypoint. The second branch predicts Part Affinity Fields that encode the degree of association between keypoints. A bipartite matching algorithm uses the Part Affinity Fields to associate keypoints into skeleton structures. This two-branch approach enables the system to handle challenging cases such as crowded scenes and partial occlusion that would confound simpler detection approaches. However, OpenPose does not incorporate exercise-specific biomechanical evaluation or provide corrective feedback, treating pose estimation as an end rather than a means to coaching applications.

MediaPipe Pose, developed by Bazarevsky et al., extended pose estimation capability to mobile devices through a lightweight architecture optimized for real-time inference on resource-constrained hardware [6]. The system predicts 33 keypoints including more detailed hand and foot landmarks than earlier approaches, providing enhanced resolution for applications requiring detailed extremity tracking. The model architecture uses efficient convolutional operations and quantization-aware training to achieve real-time performance on mobile GPUs and even CPU inference at reduced frame rates. This efficiency makes MediaPipe attractive for consumer applications where computational resources are limited.

While MediaPipe's efficiency is impressive, the fixed pre-trained weights limit adaptability to domain-specific movements. A model trained primarily on general activities may not perform optimally on the specific postures encountered in athletic training contexts. Sports movements often involve extreme ranges of motion, unusual viewing angles, and specific body configurations that may be underrepresented in general training data. The present research addresses this limitation through custom training of YOLOv8 models on exercise-specific data, enabling adaptation to the unique requirements of athletic training contexts while maintaining the inference efficiency required for real-time operation.

AlphaPose, proposed by Fang et al., achieved state-of-the-art accuracy on benchmark datasets through a regional multi-person pose estimation approach [7]. The system first detects human bounding boxes using an object detector, then performs pose estimation within each detected region. This two-stage approach improves accuracy for challenging cases involving occlusion and unusual poses by allowing the pose estimator to focus on individual people rather than the entire image. The regional attention mechanism further refines keypoint predictions by incorporating global context. However, AlphaPose requires substantial computational resources that make it unsuitable for real-time coaching applications on consumer hardware.

The processing latency introduced by AlphaPose's multi-stage pipeline exceeds the timing requirements for immediate feedback during exercise. The bounding box detection, regional pose estimation, and refinement stages each add processing time that accumulates to frame processing times often exceeding one second on typical hardware. While this latency is acceptable for offline analysis, it precludes the real-time feedback necessary for effective exercise coaching. The present research prioritizes processing efficiency to enable immediate feedback, accepting modest reductions in accuracy compared to the most complex approaches in exchange for real-time capability.

Custom trained pose models have demonstrated improved accuracy in specific domains, including clinical gait analysis and sports performance monitoring [8]. Bridgeman et al. showed that domain-specific training data and task-specific model architectures can significantly improve keypoint detection accuracy for targeted applications compared to general-purpose pose estimation models. This finding supports the approach taken in the present research, where custom-trained YOLOv8 models are developed specifically for exercise classification and pose estimation in athletic training contexts. The ability to fine-tune models on domain-specific data represents a significant advantage over fixed pre-trained approaches.

Figure 2.1 presents a taxonomy of pose estimation approaches in the literature, organizing them by architecture type, computational requirements, and application focus. The taxonomy highlights the trade-offs between accuracy, speed, and

specialization that characterize different approaches. The present research targets the intersection of real-time performance, exercise-specific accuracy, and consumer hardware compatibility, a combination not fully addressed by existing approaches.

Figure 2.1: Taxonomy of Pose Estimation Approaches in Literature Source: Author

Table 2.2 summarizes the key pose estimation approaches, comparing their architectures, keypoint counts, processing speeds, and limitations. The comparison reveals that while many approaches achieve high accuracy for general pose estimation, none are optimized for the specific requirements of exercise coaching. The present research addresses this gap through custom model training and integration with exercise-specific biomechanical evaluation.

Table 2.2: Summary of pose estimation approaches

System	Keypoints	Speed	Limitation
OpenPose [5]	18	Real-time (GPU)	No exercise evaluation
MediaPipe [6]	33	Real-time (mobile)	Fixed weights
AlphaPose [7]	17+	Slow	High computation
NeuroMotion	33	Real-time (CPU)	2D only

2.3 Exercise Recognition Systems

Deep learning approaches to exercise classification and repetition counting have shown promising results in controlled settings. Researchers have employed various neural network architectures to classify exercise types from video input or sensor data, achieving high accuracy on curated datasets. The proliferation of fitness tracking applications has created commercial interest in automated exercise logging, driving research investment in this area. However, a critical distinction exists between exercise recognition and exercise quality assessment that limits the applicability of these approaches for posture correction applications.

The fundamental challenge for exercise recognition systems is the gap between recognizing what exercise is being performed and assessing how well it is being performed. These are fundamentally different tasks requiring different data

representations and learning objectives. Exercise recognition requires learning characteristic movement patterns that distinguish one exercise type from another, such as the hip-dominant pattern of a deadlift versus the knee-dominant pattern of a squat. Quality assessment requires learning the fine-grained differences between correct and incorrect execution of the same exercise, which may involve subtle joint angle differences or timing variations that are not captured by exercise-type classifiers.

Ignatov demonstrated that convolutional neural networks can classify human activities from accelerometer data with accuracy exceeding 95% on benchmark datasets [9]. The approach used raw accelerometer signals as input to a CNN that learned discriminative features for activity classification. The network architecture included multiple convolutional layers followed by pooling and fully connected layers, trained end-to-end on labeled activity data. The high classification accuracy demonstrates the effectiveness of deep learning for recognizing distinct activity types from sensor data. While the classification accuracy is impressive, the system functioned purely as an activity recognizer without any capability to assess whether the performed activities were executed with proper form.

The Ignatov study exemplifies a common pattern in exercise recognition research: the focus on distinguishing between exercise types rather than evaluating movement quality. This focus is understandable given the commercial application to automatic exercise logging for fitness applications, where the primary requirement is identifying what exercise was performed and counting repetitions. However, this approach misses the opportunity to provide the quality feedback that would be most valuable for injury prevention and training optimization.

Pestana et al. proposed a deep learning pipeline specifically designed for counting repetitions in resistance exercises [10]. Their system used computer vision to track body keypoints through the repetition cycle, then applied temporal analysis to count complete repetitions. The approach extracted keypoints using a pose estimation model, tracked keypoint trajectories over time, and detected repetition boundaries using cyclic patterns in the trajectory data. The authors achieved high counting accuracy across

multiple exercise types including squats, push-ups, and lunges, demonstrating that computer vision can effectively track movement progression through exercise cycles.

However, Pestana et al. explicitly acknowledged that their system could not distinguish between properly and improperly executed movements. A squat performed with severe knee valgus would be counted identically to one performed with perfect form. The repetition counting system would register both as successful squats, potentially reinforcing improper technique rather than correcting it. This limitation is not specific to the Pestana et al. system but represents a broader gap in exercise recognition research that the present research addresses.

The gap between exercise recognition and posture evaluation represents a significant limitation of existing systems. Knowing that an athlete performed ten squats provides no insight into whether those squats were biomechanically sound or potentially injurious. From a training perspective, this information is of limited value; performing many repetitions with poor form is counterproductive and potentially harmful compared to performing fewer repetitions with correct form. The quality of movement execution matters at least as much as the quantity of repetitions completed, yet most existing systems ignore quality entirely.

The present research addresses this gap by integrating exercise classification with joint-level posture evaluation and corrective feedback. The system first identifies the exercise being performed, then applies exercise-specific biomechanical thresholds to evaluate the quality of execution. This integration transforms exercise recognition from an end in itself into a component of a comprehensive coaching system. The classification informs which thresholds to apply, and the threshold evaluation provides the quality assessment missing from pure recognition systems.

2.4 Voice-Based Feedback Systems

Voice-based feedback represents a natural modality for delivering coaching cues during physical activity. When athletes are actively exercising, they cannot continuously monitor a screen for visual feedback. The head and gaze are engaged in maintaining balance and coordination, and looking away from the movement space

can itself introduce form errors. Audio feedback allows athletes to receive guidance while maintaining their focus on movement execution. This principle is well-established in sports coaching, where verbal cues are a primary tool for real-time technique correction.

The effectiveness of verbal feedback in coaching contexts has been extensively studied in motor learning research. Verbal cues can direct attention to specific body parts or movement qualities, correct errors in real-time, and reinforce proper technique without interrupting the flow of exercise. The immediacy of verbal feedback enables correction before improper patterns are reinforced through repetition. These findings from coaching science provide strong justification for incorporating voice-based feedback into automated training systems.

Text-to-Speech (TTS) synthesis has advanced dramatically in recent years, transitioning from robotic-sounding early systems to natural-sounding neural voice synthesis. Early concatenative TTS systems assembled speech from recorded speech segments, producing intelligible but often unnatural output. Parametric TTS improved naturalness by generating speech from learned statistical models. Neural TTS systems, using architectures such as Tacotron and WaveNet, produce speech that is nearly indistinguishable from human voices, with appropriate prosody, emphasis, and natural variations.

These advances make voice synthesis viable for applications requiring frequent spoken output, such as exercise coaching where multiple corrections may be needed during a training session. The naturalness of neural TTS output is particularly important for applications like coaching, where an artificial-sounding voice could undermine user confidence and engagement. Modern TTS systems can generate speech in real-time on consumer hardware, enabling immediate spoken feedback without perceptible delay.

Multilingual support adds another dimension to voice feedback systems. In multilingual societies like Sri Lanka, athletes may prefer coaching in their native language rather than in English. Language preference affects not only comprehension but also emotional connection and trust. Feedback delivered in a language the athlete

prefers is more likely to be acted upon effectively. Supporting Tamil and Sinhala voice synthesis addresses a gap in existing fitness technology, which predominantly serves English-speaking users.

The technical implementation of multilingual TTS requires either pre-recorded voice libraries or neural synthesis systems trained on target languages. Browser-native Web Speech API provides access to system voices in multiple languages, offering a practical solution for web-based applications without requiring additional dependencies. The Web Speech API abstracts over platform-specific speech synthesis implementations, providing a consistent programming interface across different browsers and operating systems. However, voice quality and language availability vary by platform, which can affect user experience.

The present research integrates multilingual voice feedback with biomechanical analysis, creating a feedback channel that operates in parallel with visual feedback. When posture errors are detected, the system generates spoken corrective instructions in the athlete's preferred language. This integration addresses the practical limitation of visual feedback during active exercise while also providing language accessibility for diverse user populations.

2.5 Summary

The literature review reveals that existing systems address isolated components of exercise monitoring but no integrated solution combines all necessary capabilities for effective real-time posture correction. Wearable systems provide movement data but cannot assess joint-level posture due to fundamental limitations of inertial sensing. Motion capture laboratories provide accurate analysis but require expensive equipment and controlled environments that limit accessibility. Computer vision systems can extract pose information but typically do not provide exercise-specific evaluation or corrective feedback. Exercise recognition systems count repetitions without assessing quality, missing the most important dimension for training optimization. Voice feedback systems exist but have not been integrated with biomechanical analysis in a multilingual context.

This research addresses these gaps through an integrated approach that combines camera-only pose estimation with exercise-specific biomechanical thresholds and multilingual voice feedback. By leveraging recent advances in deep learning, particularly the YOLOv8 architecture with its native support for both classification and keypoint estimation tasks, the present work achieves professional-grade posture analysis using standard consumer hardware. The following chapter describes the methodology developed to realize this integrated approach, including system architecture, mathematical formulations, and implementation details.

3. METHODOLOGY

3.1 System Architecture Overview

NeuroMotion operates as a browser-based Python application built with the Streamlit framework, designed to provide real-time biomechanical exercise correction through an intuitive web interface. The system architecture follows a modular design pattern that separates concerns across distinct processing stages, enabling efficient data flow from video capture through feedback delivery. This architectural approach facilitates maintenance, testing, and future extension of individual components without affecting the overall system integrity.

The system supports three operational modes to accommodate different training scenarios. The single player live training mode utilizes a connected camera for real-time exercise analysis, providing immediate feedback during active training sessions. The video upload analysis mode enables post-session review, allowing athletes to analyze previously recorded training sessions for detailed feedback. The 1v1 multiplayer competitive mode introduces a gamified element where two athletes compete while receiving form feedback, enhancing engagement and motivation for correct technique. Each mode shares the same underlying processing pipeline but presents results differently based on the use case.

Figure 3.1 illustrates the complete system architecture spanning from input capture through user interaction layers. The processing pipeline begins with video input, either from a live camera stream or an uploaded file. This input passes through two sequential YOLOv8 models: the first model classifies the exercise being performed, while the second model estimates the 33 body keypoints that define the skeletal pose. The detected keypoints then flow through the joint angle calculation module, which computes the biomechanically relevant angles for the detected exercise type.

Figure 3.1: NeuroMotion System Architecture Pipeline Source: Author

All backend processing occurs on the local machine, ensuring real-time responsiveness without network dependency. This design choice eliminates latency introduced by cloud processing and maintains functionality even in environments with limited or no internet connectivity. The local processing architecture also addresses privacy concerns by keeping video data on the user's device rather than transmitting it to external servers. The Streamlit framework provides the web interface layer, rendering results in real-time to the browser while the Python backend handles all computational work.

3.2 Human Body Modeling for Pose Estimation

To accurately capture athletic movements, the human body is represented as a skeletal model comprising 33 anatomically significant keypoints arranged in a hierarchical tree structure. This representation draws from the COCO keypoint format but extends it to capture finer details necessary for exercise analysis. The additional keypoints beyond the standard 17-point COCO model provide enhanced resolution at extremities and facial features, supporting more nuanced posture evaluation.

The skeletal topology consists of five major body segments organized by functional anatomy. The head region includes 5 keypoints representing the nose, left eye, right eye, left ear, and right ear. While these facial landmarks are detected, they serve primarily for reference and head orientation estimation rather than direct posture evaluation. The upper torso segment comprises the left and right shoulders paired with the left and right hips, forming the core stability frame that governs overall body alignment during most exercises.

The upper extremities segment captures arm movement through keypoints at the elbows, wrists, and hands. This segment is particularly important for exercises involving arm positioning, such as push-ups where elbow angle determines proper depth and hand placement relative to the shoulders. The lower extremities segment includes keypoints at the knees, ankles, and feet, enabling detailed analysis of leg mechanics during squats, jumps, and other lower-body exercises. Each keypoint stores

its 2D pixel coordinates (x, y) along with a confidence score ranging from 0 to 1, indicating the detection reliability.

Figure 3.2 presents the complete 33-keypoint skeletal model with anatomically labeled landmarks. The model structure follows biomechanical conventions where keypoints correspond to anatomically significant joints and reference points. The connectivity between keypoints defines the skeletal segments that will be analyzed for joint angles and overall posture quality.

Figure 3.2: 33-Keypoint Human Skeletal Model with Labeled Keypoints Source: Author

The modeling approach differs fundamentally from general-purpose pose estimation in several ways. First, the system prioritizes biomechanically relevant joints over aesthetic completeness. Greater computational emphasis is placed on hip-knee-ankle alignment for squat evaluation and shoulder-elbow-wrist trajectories for push-up assessment. Second, temporal consistency is enforced through Exponential Moving Average (EMA) smoothing applied to keypoint positions across consecutive frames.

Given a keypoint position at frame t denoted as P(t), the smoothed position P'(t) is calculated using the recursive formula:

$$P'(t) = \alpha \times P(t) + (1 - \alpha) \times P'(t - 1)$$

Where α represents the smoothing factor set to 0.7 based on empirical tuning. This value provides a balance between responsiveness to actual movement and suppression of high-frequency detection noise. Lower values of α would introduce lag, making the system feel sluggish, while higher values would fail to filter out jitter from detection instability.

The skeletal model incorporates kinematic chain constraints that reflect human anatomical limitations. The knee joint is modeled as a hinge joint with a valid flexion range of 0° to 140°, while the hip joint operates as a ball-and-socket joint with a broader range of motion. These constraints are enforced during posture evaluation to reject physiologically impossible keypoint configurations that may arise from occlusion or detection errors.

3.3 Dataset Collection and Preparation

The training dataset comprised over 500 labeled exercise video clips sourced from multiple origins to ensure diversity and generalization capability. The multi-source approach addresses the common machine learning challenge of dataset bias, where models trained on homogeneous data fail to generalize to varied real-world conditions. By combining public datasets with custom-recorded footage, the system achieves robustness across different camera types, lighting conditions, and subject demographics.

The COCO Keypoints dataset contributed 250,000 person instances with 17 keypoint annotations that were extended to the 33-keypoint format required by the pose model [12]. The COCO dataset provides diverse images capturing people in various poses, activities, and environments. While the original annotations include only 17 keypoints, the additional 16 keypoints required for the extended format were synthesized using anatomical heuristics and validated through manual review of a subset of instances.

The Kaggle Exercise Dataset provided labeled video clips of squats, push-ups, and jumps captured under varied conditions. This dataset contributed domain-specific examples of the target exercises, supplementing the general human pose data from COCO with movement patterns specific to athletic training. The videos included variations in execution style, from technically correct form to common errors, providing the model with exposure to the range of movements it would need to evaluate in deployment.

Custom footage was recorded from over 20 volunteers at 720p resolution and 30 frames per second (FPS) using standard webcams. This data collection effort deliberately varied lighting conditions from bright outdoor settings to dimly lit indoor environments, body types across different heights and body compositions, camera angles including front-facing, side-facing, and oblique views, and clothing styles to ensure the model is not biased toward specific athletic wear. The diversity in the custom dataset addresses the practical deployment scenario where users will exercise in varied home or gym environments with available cameras.

Table 3.1 presents the dataset composition and the train/validation/test split used for model development. The split ratio of 70% training, 15% validation, and 15% testing follows standard machine learning practice, ensuring sufficient data for model training while reserving adequate held-out sets for hyperparameter tuning and final performance evaluation.

Table 3.1: Dataset composition and train/validation/test split

Dataset Component	**Quantity**	**Percentage**
Training set	350+ videos	70%
Validation set	75+ videos	15%
Testing set	75+ videos	15%
Total	500+ videos	100%
Custom footage volunteers	20+ subjects	-
COCO person instances	250,000	-

Data augmentation techniques applied during training enhanced the model's robustness to variations in input. Random horizontal flips introduced bilateral symmetry exercises, effectively doubling the training data for symmetric movements. Brightness and contrast variations of ±30% simulated different lighting conditions. Random scaling between 0.8× and 1.2× accommodated different distances from the camera. Perspective warping simulated diverse camera angles that users might employ during exercise sessions.

3.4 YOLOv8 Model Customization and Training

The YOLOv8 architecture was adapted for the dual-task requirement of exercise classification and pose estimation through a customized multi-head configuration. Unlike previous YOLO versions that required separate models for different tasks, YOLOv8 natively supports both classification and keypoint estimation within a unified training framework, reducing integration complexity and enabling joint optimization of both tasks [13].

The base network employs the CSPDarknet53 backbone for feature extraction, followed by a Feature Pyramid Network (FPN) that processes multi-scale features at three resolution levels: 80×80, 40×40, and 20×20 pixels. This pyramid structure enables detection of keypoints at varying distances from the camera while maintaining computational efficiency suitable for real-time operation. The FPN architecture addresses the challenge that body parts can appear at different scales depending on the subject's distance from the camera and their position within the frame.

For the classification task, the network head was modified to output a 4-dimensional probability distribution corresponding to the exercise classes: squat, push-up, jump, and none. The classification head consists of a global average pooling layer that collapses spatial dimensions, followed by two fully connected layers of 512 and 128 neurons respectively, with ReLU activation and dropout (rate=0.3) applied between them to prevent overfitting. The final layer uses softmax activation to produce class probabilities.

For the pose estimation task, the detection head was reconfigured to predict 33 keypoints per person instance. Each keypoint prediction comprises three components: the x-coordinate offset, the y-coordinate offset, and a confidence score. The pose head branches from the FPN features and applies a series of 3×3 convolutional layers with 256 channels, progressively refining the keypoint heatmaps. The final prediction layer outputs a tensor of shape (batch, 33, 3) representing all keypoint coordinates and confidences.

Figure 3.3 presents the complete dual-model pipeline architecture, illustrating the flow from input video through exercise classification and pose estimation to the final output predictions.

Figure 3.3: YOLOv8 Dual-Model Pipeline Architecture Source: Author

Transfer learning was implemented by initializing the backbone weights from ImageNet pre-trained parameters, leveraging the rich feature representations learned from millions of natural images. The pose estimation branch was initialized from COCO-trained weights, providing a strong starting point for human keypoint

detection. This transfer learning approach significantly reduces training time and improves final model performance compared to training from scratch.

The training process employed a two-phase strategy to optimize both stability and performance. In the first phase (epochs 1-50), only the classification and pose heads were trained while the backbone remained frozen with its pre-trained weights. This phase allows the task-specific layers to adapt without disrupting the general feature extraction capabilities of the backbone. In the second phase (epochs 51-150), the entire network was fine-tuned with a reduced learning rate of 1×10^{-4}, enabling the backbone to specialize for exercise-specific visual patterns while maintaining the improvements from the first phase.

The total loss function combines classification cross-entropy, pose regression, and confidence prediction terms with empirically determined weights that balance their contributions. The loss function is defined as:

$$L_total = \lambda_1 \times L_cls + \lambda_2 \times L_pose + \lambda_3 \times L_conf$$

Where $\lambda_1 = 0.3$, $\lambda_2 = 0.5$, and $\lambda_3 = 0.2$ represent weights determined through validation set performance evaluation. The pose loss L_pose uses an L1 distance between predicted and ground truth keypoint coordinates, providing robust gradient signals for localization. The classification loss L_cls applies standard cross-entropy for multi-class classification. The confidence loss L_conf uses binary cross-entropy for keypoint visibility classification.

3.5 Joint Angle Calculation

Joint angles constitute the core quantitative measurements used for posture evaluation. The calculation method employs vector-based trigonometry to compute the angle formed at a joint by two connected body segments. This approach provides biomechanically meaningful quantities that can be compared against established thresholds for correct form, enabling objective and consistent posture assessment.

For three connected keypoints A, B, and C, where B represents the joint of interest, the angle at B is calculated using the two-argument arctangent (atan2)

function. The atan2 function is preferred over the single-argument arctangent because it returns angles in the correct quadrant based on the signs of its arguments, avoiding ambiguity in angle determination. Given the 2D coordinates A = (x_A, y_A), B = (x_B, y_B), and C = (x_C, y_C), two vectors are constructed:

$$Vector\ BA = (x_A - x_B, y_A - y_B) \quad [from\ B\ pointing\ to\ A]$$

$$Vector\ BC = (x_C - x_B, y_C - y_B) \quad [from\ B\ pointing\ to\ C]$$

The raw angle θ at joint B is computed using the atan2 function as follows:

$$\theta = atan2(BC_y, BC_x) - atan2(BA_y, BA_x)$$

Since the raw result may fall outside the range of valid joint angles, normalization is applied to obtain the interior angle:

$$\theta_normalized = min(|\theta|, 360° - |\theta|)$$

This normalization ensures the angle is always the smaller interior angle at the joint, which corresponds to the biomechanically relevant flexion angle. For example, a straight leg segment from hip to knee to ankle produces an angle approaching 180°, while a deeply flexed knee might measure around 90° or less.

Figure 3.4 provides a geometric visualization of the three-keypoint configuration, showing how vectors BA and BC are constructed from joint center B to compute the interior angle θ at the target joint.

Figure 3.4: Joint Angle Calculation Geometry Showing Keypoints A, B, C and Angle θ Source: Author

For knee valgus detection specifically, the angle formed by the hip-knee-ankle triplet is compared against the ideal alignment of 180°. The deviation is calculated as:

$$\Delta_valgus = |180° - \theta_knee|$$

A deviation exceeding 15° triggers a valgus error, indicating that the knee is deviating medially from the straight line connecting hip and ankle. Similarly, forward lean is assessed by computing the shoulder-hip-knee angle, with values below 150° indicating excessive forward inclination that places undue stress on the lumbar spine.

3.6 Posture Evaluation and Biomechanical Thresholds

Exercise-specific biomechanical thresholds were established based on sports medicine literature and empirical testing to enable meaningful posture evaluation. These thresholds define the acceptable ranges for joint angles and body positions that characterize correct form for each supported exercise. The evaluation logic compares computed joint angles against these thresholds, triggering appropriate error classifications when violations are detected.

For squats, the system implements four distinct posture checks. Knee valgus detection triggers an error when the deviation from straight alignment exceeds 15°. Forward lean assessment flags an error when the shoulder-hip-knee angle falls below 150°. Knee over toes detection identifies when the knee x-coordinate exceeds the ankle x-coordinate by more than 10% of frame width. Back arch detection triggers when the shoulder-hip angle drops below 160°.

For push-ups, the system implements four posture checks adapted to the horizontal body orientation. Back arch detection uses the same threshold as squats but evaluates spine alignment in the prone position. Head position assessment flags an error when the nose exceeds 10% of frame height from shoulder alignment. Arm angle evaluation checks whether the elbow angle falls within the acceptable range of 90° to 150°. Hip sag detection triggers when the hip position drops more than 10% below the shoulder-ankle reference line.

For jumps, the system implements three posture checks focused on landing mechanics. Knee valgus and forward lean assessments use the same thresholds as squats. The knee over toes check ensures proper landing alignment with the knees tracking over rather than past the toes.

Table 3.2 presents the complete set of biomechanical thresholds organized by exercise type and error category. Each threshold value is accompanied by the biomechanical rationale derived from sports medicine literature.

Figure 3.5 illustrates the four error types evaluated during squat analysis, providing a visual representation of each biomechanical deviation and its associated detection threshold.

Table 3.2: Exercise-specific biomechanical thresholds

Exercise	Error Type	Threshold	Condition
Squat	Knee Valgus	$\Delta > 15°$	$\lvert 180° - \theta_knee \rvert > 15°$
Squat	Forward Lean	$\theta < 150°$	Shoulder-hip-knee angle
Squat	Knee Over Toes	> 10% frame	Knee X > Ankle X + 10%
Squat	Back Arch	$\theta < 160°$	Shoulder-hip angle
Push-up	Back Arch	$\theta < 160°$	Spine alignment
Push-up	Head Position	> 10% frame	Nose from shoulder line
Push-up	Arm Angle	90°–150°	Elbow angle range
Push-up	Hip Sag	> 10% below	Hip below reference
Jump	Knee Valgus	$\Delta > 15°$	Same as squat
Jump	Forward Lean	$\theta < 150°$	Same as squat
Jump	Knee Over Toes	> 10% frame	Same as squat

Figure 3.5: Biomechanical Posture Error Types for Squat Exercise Evaluation Source: Author

The scoring system incentivizes correct form over high repetition counts. Each completed repetition earns 10 base points, while each unique postural error detected in that repetition results in a 2-point deduction. This scoring model encourages athletes to focus on movement quality rather than simply accumulating repetitions.

3.7 Visual Feedback: Skeleton Overlay and Guideline Lines

The visual feedback system provides immediate, intuitive indication of posture quality through color-coded skeleton overlays and three horizontal guideline lines. This dual-channel visual feedback enables athletes and coaches to quickly identify both specific joint-level errors and overall body alignment issues during exercise performance.

The skeleton overlay renders the 33 detected keypoints connected as a skeletal structure superimposed on the video feed. The color coding provides instant posture status indication: green skeleton indicates all posture checks pass, representing correct form, while red skeleton signals that at least one posture check has failed. This binary color distinction allows athletes to know immediately whether their current form is acceptable or requires correction.

Three horizontal guideline lines provide additional spatial reference for body alignment assessment. The yellow line serves as the Reference Stability Line, positioned to evaluate vertical balance and overall body position. The blue line functions as the Dynamic Movement Tracking Line, changing position depending on the exercise to follow major movement regions. The green line represents the Core Body Alignment Line, positioned around the hip and central body region. Figure 3.6 demonstrates the three guideline lines displayed on the skeleton overlay during a squat exercise, showing the spatial positioning of the yellow stability line, the dynamic blue tracking line, and the green core alignment line.

Figure 3.6: Three Guideline Lines (Yellow, Blue, Green) on Skeleton Overlay Source: Author

3.8 Voice-Based Corrective Feedback Module

The voice-based corrective feedback module delivers spoken guidance to athletes when posture errors are detected. This modality addresses a fundamental constraint of visual feedback: athletes actively exercising cannot continuously monitor a screen. Audio feedback enables athletes to receive corrections while maintaining their focus on movement execution, creating a coaching experience that mirrors the verbal cues provided by human trainers.

The feedback system generates contextually appropriate corrective instructions based on the specific error detected. For knee valgus, the system advises keeping the knees in line with the toes. For forward lean, the system suggests maintaining an upright torso. Each error type maps to a specific instructional phrase designed to guide the athlete toward correct form without requiring technical knowledge of biomechanics.

Rate limiting prevents overwhelming the athlete during periods of multiple concurrent errors. The system enforces a minimum interval of 4 seconds between spoken corrections. If multiple errors occur within this window, only the first detected error triggers voice feedback, with subsequent errors queued for the next available feedback opportunity.

Figure 3.7 presents the complete flow of the voice feedback system, from posture error detection through language selection to audio delivery, including the rate limiting logic.

Figure 3.7: Voice Feedback System Flow Diagram Source: Author

3.9 Multilingual Text-to-Speech Implementation

The multilingual voice feedback system supports three languages: English, Tamil, and Sinhala. This language accessibility addresses a gap in existing fitness technology, which predominantly serves English-speaking users. In the Sri Lankan context, many athletes prefer coaching and instruction in their native language, making multilingual support essential for adoption and effectiveness.

The implementation utilizes the Web Speech API, a browser-native interface for speech synthesis that provides access to system voices in multiple languages. This approach avoids dependency on external services, maintaining the system's local processing architecture. Language selection is configurable through the user interface, allowing athletes to choose their preferred feedback language.

Table 3.3 presents the complete technology stack used in the NeuroMotion system, listing each architectural layer alongside the specific technology selected for implementation.

Table 3.3: Technology stack of the NeuroMotion system

Layer	Technology
Backend Language	Python 3.10+
Frontend Framework	Streamlit
AI / Deep Learning	Ultralytics YOLOv8

Layer	Technology
Computer Vision	OpenCV
Numerical Computing	NumPy
Machine Learning	scikit-learn
Data Visualization	Plotly, Pandas
Database	SQLite
Authentication	bcrypt
Voice Synthesis	Web Speech API

3.10 Commercialization Aspects of the Product

The NeuroMotion system addresses a significant market opportunity in the sports training and fitness technology sector. The target market encompasses sports academies seeking to provide consistent coaching across all athletes, university athletics programs with limited coaching resources, commercial gyms looking to differentiate their offerings, individual personal trainers who can extend their reach through technology, and rehabilitation centers requiring objective movement assessment.

The proposed business model follows a Software as a Service (SaaS) subscription framework with tiered pricing. Sports academies would subscribe to institutional plans enabling deployment across multiple training stations. Individual athletes would access the system through freemium or subscription models. The competitive advantage stems from the unique combination of camera-only operation, multilingual support, and real-time AI coaching with exercise-specific biomechanical thresholds.

3.11 Testing and Implementation

The testing methodology encompassed both algorithmic validation and user experience evaluation. Algorithmic testing evaluated posture detection accuracy against expert-annotated ground truth labels. The test set comprised 75 video clips containing multiple exercise repetitions performed by subjects not seen during

training. Three independent human annotators labeled posture quality for each frame, with inter-annotator agreement of 97.2% establishing reliable ground truth.

Performance testing measured frame processing latency under various conditions. The sub-100ms frame processing latency on CPU-only systems ensures that visual correction appears within approximately three frames at 30 FPS, well within the human perceptual window for real-time response. This latency is sufficiently fast for athletes to adjust mid-movement.

Figure 3.8 summarizes the two-phase training strategy, illustrating the transition from frozen backbone training in Phase 1 (epochs 1–50) to full network fine-tuning in Phase 2 (epochs 51–150).

Figure 3.8: Two-Phase YOLOv8 Training Strategy Diagram Source: Author

3.11.1 Functional and Non-Functional Requirements

FR/NFR	ID	Requirement	Status
Functional	FR1	System shall detect squat posture in real time	Met
Functional	FR2	System shall classify exercise type with ≥90% accuracy	Met
Non-Functional	NFR1	Frame processing latency shall be < 100ms	Met

4. RESULTS AND DISCUSSION

4.1 System Performance Results

The evaluation of the NeuroMotion system focused on six key performance metrics that together characterize the system's effectiveness for real-time exercise correction. Table 4.1 presents the overall performance evaluation results, demonstrating strong performance across all measured dimensions. These results validate the technical approach and confirm that the system meets the requirements for practical deployment in training environments.

Table 4.1: Overall performance evaluation results

Metric	Value
Posture Detection Accuracy	92.1%
Exercise Classification Accuracy	≥ 90%
Repetition Counting Accuracy	94.8%
Frame Processing Latency (CPU)	< 100ms
Voice Alert Delivery Rate	≥ 98%
System Calibration Time	3.3 seconds
System Uptime (30-min session)	No crashes

The 92.1% posture detection accuracy represents frame-level agreement between the system's posture verdict and expert-annotated ground truth labels across the held-out test set of 75 video clips. Each clip contained multiple exercise repetitions performed by subjects not seen during training, ensuring the accuracy metric reflects generalization capability rather than memorization of training examples. This accuracy level exceeds the 89% achieved by comparable wearable-based systems while providing joint-level posture assessment that those systems cannot offer.

The sub-100ms frame processing latency ensures that visual correction appears within approximately three frames at 30 FPS, well within the human perceptual window for real-time response. This latency is sufficiently fast for athletes to adjust mid-movement, enabling corrective action during the exercise rather than post-hoc

analysis. The measurement was taken on a standard laptop without GPU acceleration, indicating that the system meets real-time requirements even on consumer hardware.

Repetition counting accuracy of 94.8% was validated by having three independent human counters watch the same sessions and comparing their tallies against system output. Inter-annotator agreement among the human counters measured 97.2%, establishing reliable ground truth for comparison. The counting accuracy demonstrates the system's ability to correctly identify repetition boundaries through temporal analysis of posture transitions.

The voice alert delivery rate of ≥98% confirms the reliability of the speech synthesis system. Failed deliveries were primarily attributable to browser audio system conflicts during initial testing, which were resolved through implementation improvements. The 3.3-second calibration period, capturing baseline posture through a 100-frame standing reference, was found in user testing to be non-intrusive.

4.2 Posture Detection Accuracy by Exercise Type

Exercise-specific detection rates reveal performance patterns tied to movement characteristics. Table 4.2 presents the posture detection accuracy broken down by exercise type, showing the number of posture checks evaluated for each exercise and the achieved accuracy.

Table 4.2: Posture detection accuracy by exercise type

Exercise	Posture Checks	Accuracy
Squat	4	92.3%
Push-up	4	91.1%
Jump	3	92.8%

Jump analysis achieves the highest accuracy at 92.8%, benefiting from a cleaner mid-air posture window when both feet leave the ground. During this phase, upper body sway is reduced, giving the model a clearer keypoint signal with fewer occlusions. The relatively simpler posture evaluation, with only three checks compared to four for squats and push-ups, also contributes to the higher accuracy.

Squat detection maintains 92.3% accuracy despite monitoring four distinct posture parameters simultaneously. The system must evaluate knee valgus deviation, forward lean angle, knee-over-toes alignment, and back arch in combination. Maintaining accuracy across all four constraints demonstrates the robustness of the angle-based evaluation approach.

Push-up analysis shows slightly lower accuracy at 91.1%, attributable to upper body joint tracking challenges. Wrist and elbow keypoints are occasionally occluded when the user's body is close to the floor, which introduces landmark detection uncertainty during the lowest phase of the movement. Future work will address this limitation through depth-aware keypoint confidence weighting. Figure 4.1 presents a visual comparison of posture detection accuracy across the three supported exercises, illustrating the relative performance differences and their relationship to exercise complexity and keypoint visibility.

Figure 4.1: Posture Detection Accuracy Comparison Chart by Exercise Type Source: Author

4.3 Voice Feedback System Evaluation

The voice feedback system was evaluated across three dimensions: technical reliability, language quality, and user acceptance. Technical reliability was measured by the voice alert delivery rate, which exceeded 98% across all test sessions. Language quality assessment involved native speakers of each supported language evaluating the naturalness and clarity of corrective phrases.

English voice synthesis achieved near-native quality using standard browser voices available across platforms. Tamil voice synthesis quality varied depending on the browser and operating system, with some platforms providing more natural voices than others. Sinhala voice synthesis faced the greatest challenge due to limited availability of high-quality Sinhala voices in browser speech engines.

The rate limiting strategy of one correction per 4 seconds received positive feedback from participants. This interval was described as allowing sufficient time to process the feedback and attempt correction without feeling overwhelmed. Several

participants noted that faster feedback would have been difficult to process while maintaining exercise rhythm.

4.4 Comparison with Existing Systems

Comparing NeuroMotion against existing exercise analysis solutions highlights its unique combination of capabilities. Table 4.3 presents a feature-by-feature comparison across four alternative approaches: wearable sensor systems, MediaPipe-based applications, motion capture laboratories, and the proposed NeuroMotion system.

Table 4.3: Comparison with existing approaches

Feature	**Wearable**	**MediaPipe**	**Motion Capture**	**NeuroMotion**
Exercise-specific corrective feedback	No	No	No	Yes
Camera only (no wearables needed)	No	Yes	No	Yes
Joint angle analysis	Partial	No	Yes	Yes
Exercise-specific thresholds	No	No	Yes	Yes
Multilingual voice feedback	No	No	No	Yes
Performance dashboard	No	Partial	Yes	Yes
Affordable for individual athletes	No	Yes	No	Yes

Wearable sensor systems lack the skeletal understanding necessary for joint-level posture assessment. MediaPipe-based applications offer camera-only operation but provide no exercise-specific thresholds or real-time corrective feedback. Motion capture laboratories deliver highly accurate analysis but require controlled environments and prohibitively expensive equipment.

The custom trainable YOLOv8 architecture enables domain adaptation for new exercises without infrastructure changes. As the exercise library expands to include additional movements, the same training pipeline can incorporate new exercise classes and biomechanical thresholds. Unlike preceding architectures such as YOLOv7 [11],

YOLOv8 natively supports both classification and keypoint estimation tasks within a unified training pipeline.

4.5 Research Findings

The research findings can be organized into three categories: technical achievements, user experience insights, and broader implications for accessible biomechanical coaching. The technical achievements demonstrate that real-time biomechanical exercise correction using computer vision is feasible on consumer hardware with accuracy levels previously achievable only through expensive specialized equipment.

The joint angle calculation methodology using vector trigonometry proved accurate and computationally efficient. The atan2-based approach handles all joint configurations correctly without the quadrant ambiguity issues that affect simpler arctangent-based methods. The normalization to interior angles provides biomechanically meaningful quantities that directly map to established thresholds in sports medicine literature.

Figure 4.2 presents a radar chart comparing the system's performance across six evaluation dimensions, providing a multi-dimensional view of the overall system capability relative to target specifications.

Figure 4.2: System Performance Radar Chart Across Multiple Metrics Source: Author

4.6 Discussion

The results demonstrate that the research objectives have been successfully achieved. The system provides real-time, joint-level posture feedback using only a standard camera, addressing the primary objective of making professional-grade biomechanical coaching accessible. The 92.1% posture detection accuracy confirms the effectiveness of the technical approach, while the sub-100ms latency validates the real-time capability.

Several limitations warrant acknowledgment and consideration for future development. The system operates with monocular camera input only, performing 2D

analysis without depth estimation. This limitation may reduce accuracy for movements with significant forward-backward motion relative to the camera plane. Exercises where depth information is critical would benefit from stereo camera or depth sensor integration.

The SQLite database architecture employs single-writer semantics, making it unsuitable for concurrent multi-user deployment without migration to a server-based database system. For single-user applications, this limitation has no practical impact, but institutional deployments with multiple concurrent users would require database infrastructure changes.

4.7 Summary of My Individual Contribution

My individual contribution to the NeuroMotion project focused on the Computer Vision for Athlete Form and Technique Analysis component and the Voice-Based Corrective Feedback System. These two interrelated modules constitute the core intelligence of the system, enabling real-time posture detection and immediate corrective guidance delivery.

Specifically, I was responsible for designing and implementing the dual YOLOv8 model architecture for exercise classification and 33-keypoint pose estimation. This included the customization of network heads for our specific task requirements, the implementation of the two-phase training strategy, and the optimization of hyperparameters for real-time performance. The mathematical framework for joint angle calculation using vector trigonometry was developed and validated under my direction.

The visual feedback system, including the color-coded skeleton overlay and three guideline lines, was designed and implemented by me. The guideline line positioning logic, which adapts to different exercises and movement phases, was developed to provide intuitive spatial reference for posture assessment. The voice-based corrective feedback module, including the multilingual support for English, Tamil, and Sinhala, was also my contribution.

5. CONCLUSION

5.1 Summary of the Research

This dissertation presented NeuroMotion, a real-time biomechanical exercise correction system that demonstrates the feasibility of providing professional-grade posture coaching using only a standard camera and consumer computing hardware. The system achieves 92.1% posture detection accuracy across three common exercises, squats, push-ups, and jumps, while maintaining sub-100ms frame processing latency suitable for real-time coaching applications.

The research addressed a significant gap in existing exercise monitoring technology: the absence of an integrated solution combining camera-only operation, joint-level posture feedback, exercise-specific biomechanical thresholds, and multilingual voice coaching. While existing systems address isolated components, such as wearable-based movement detection or pose estimation without evaluation, no prior system has successfully integrated all necessary capabilities for effective real-time posture correction accessible to athletes at all competitive levels.

The technical approach leveraged recent advances in deep learning, particularly the YOLOv8 architecture with its native support for both classification and keypoint estimation tasks. The dual-model pipeline first classifies the exercise being performed, then estimates the 33 body keypoints that define skeletal configuration. Joint angles computed through vector trigonometry are evaluated against exercise-specific biomechanical thresholds to determine posture correctness.

The feedback system delivers guidance through both visual and auditory channels. The visual channel provides immediate indication of posture status through color-coded skeleton overlay and three spatial reference lines. The auditory channel delivers spoken corrective instructions in the athlete's preferred language, enabling hands-free coaching during active exercise. The multilingual support for English, Tamil, and Sinhala addresses a gap in existing fitness technology that predominantly serves English-speaking users.

5.2 Achievements and Contributions

The primary achievement of this research is the successful implementation of a complete real-time biomechanical exercise correction system that operates on consumer hardware. The system provides functionality previously available only through expensive motion capture laboratories or required multiple body-worn sensors, democratizing access to professional-grade posture coaching.

The specific technical contributions include: (1) a dual YOLOv8 model architecture optimized for the dual-task requirement of exercise classification and pose estimation; (2) a mathematical framework for joint angle calculation using vector trigonometry with appropriate normalization; (3) exercise-specific biomechanical thresholds derived from sports medicine literature for squats, push-ups, and jumps; (4) a visual feedback system with color-coded skeleton overlay and adaptive guideline lines; (5) a voice-based corrective feedback module with multilingual support; and (6) a two-phase training strategy for effective transfer learning.

The research contributes to the broader field of accessible biomechanical coaching by demonstrating that computer vision technology can bridge the gap between expensive professional systems and affordable consumer applications. The implications extend beyond athletic training to rehabilitation, elderly fall prevention, and workplace ergonomics, where accessible posture analysis could have significant health and safety benefits.

5.3 Limitations

Several limitations of the current system warrant acknowledgment. The monocular camera input restricts analysis to 2D joint angles without depth information. Movements with significant forward-backward motion relative to the camera may experience reduced accuracy due to the inability to distinguish depth from a single viewpoint. Exercises where depth is critical for proper form assessment would benefit from stereo camera or depth sensor integration.

The current exercise library is limited to three exercises: squats, push-ups, and jumps. While these represent fundamental movement patterns common across training programs, expansion to additional exercises such as lunges, deadlifts, overhead presses, and sport-specific movements would enhance the system's utility for broader training contexts.

The SQLite database architecture limits deployment scalability for multi-user institutional settings. Single-writer semantics prevent concurrent access from multiple users, requiring migration to a server-based database system for deployment in gyms, training centers, or other multi-user environments.

Voice synthesis quality for Tamil and Sinhala depends on browser and operating system support. Platforms without native voices for these languages require additional language pack installation, potentially creating friction for users who prefer these languages.

5.4 Future Work and Recommendations

Future development will focus on four primary areas: cloud deployment for scalable multi-user access, mobile application development for broader accessibility, 3D pose estimation through depth sensing, and exercise library expansion. Cloud deployment using containerized inference on platforms such as AWS or Google Cloud Run would enable sub-50ms inference latency through GPU acceleration while supporting multiple concurrent users across distributed training facilities.

A lighter mobile version using model quantization and pruning techniques would enable on-device inference for individual athletes and personal trainers. Target model size below 10MB would allow deployment on consumer smartphones without requiring continuous network connectivity. This mobile capability would extend the system's reach to outdoor training environments and personal training sessions.

Depth estimation integration through stereo camera or LiDAR sensors would enable full 3D joint angle computation, addressing the current limitation of monocular 2D-only analysis. Stereo disparity maps would provide the depth information

necessary to accurately assess forward-backward motion and distinguish posture errors that are ambiguous in 2D projection.

Exercise library expansion is planned to include lunges, deadlifts, overhead presses, and other resistance training movements identified as high injury-risk exercises in preliminary coach interviews. Each additional exercise requires definition of appropriate biomechanical thresholds, which will be developed in consultation with sports medicine professionals and validated through user testing.

In conclusion, this research has successfully demonstrated that real-time biomechanical exercise correction using computer vision is achievable on consumer hardware with accuracy levels suitable for practical training applications. The NeuroMotion system makes professional-grade posture coaching accessible to athletes who previously could not afford or access such technology, contributing to improved training effectiveness and reduced injury risk across the athletic population.

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APPENDICES

Appendix A: YOLOv8 Model Architecture Details

This appendix provides detailed specifications of the YOLOv8 model architecture used in the NeuroMotion system. The architecture follows the standard YOLOv8 structure with modifications for the dual-task requirement of exercise classification and pose estimation.

The backbone network uses CSPDarknet53 with cross-stage partial connections that improve gradient flow during training while reducing computational cost. The network consists of 53 convolutional layers organized into stages with residual connections. Each stage progressively reduces spatial resolution while increasing channel depth, enabling multi-scale feature extraction.

The Feature Pyramid Network (FPN) processes backbone features at three scales: P3 (80×80), P4 (40×40), and P5 (20×20). The FPN uses top-down pathway with lateral connections to combine high-level semantic features with low-level spatial features. This architecture enables detection of keypoints at various scales, essential for handling different camera distances and body orientations.

The classification head consists of global average pooling followed by two fully connected layers. The first FC layer has 512 neurons with ReLU activation and dropout (rate=0.3). The second FC layer has 128 neurons with ReLU activation. The final output layer produces 4-class probabilities using softmax activation.

The pose estimation head branches from FPN features and applies a series of 3×3 convolutions with 256 channels. Deconvolution layers upsample features to the target resolution before final keypoint prediction. Each keypoint prediction includes x-coordinate, y-coordinate, and confidence score.

Appendix B: Biomechanical Threshold Derivation

This appendix describes the derivation of biomechanical thresholds used for posture evaluation. The thresholds were established through review of sports medicine

literature, consultation with biomechanics experts, and empirical testing on our dataset.

Knee valgus threshold of 15° deviation from 180° was derived from research on ACL injury risk factors. Studies have consistently shown that knee valgus angles greater than 15° during landing and cutting movements significantly increase ACL loading and injury risk.

Forward lean threshold of 150° for the shoulder-hip-knee angle was established based on research on lumbar spine loading during squatting. Angles below this threshold shift load from the hip extensors to the lumbar spine, increasing injury risk.

The 10% frame width threshold for knee-over-toes was determined empirically based on anatomical proportions and camera perspective considerations. This threshold accounts for natural variation in limb proportions while identifying clinically significant anterior knee displacement.

Appendix C: User Testing Protocol and Results

User testing was conducted with 25 participants including recreational athletes, fitness enthusiasts, and personal trainers. The testing protocol involved a familiarization session followed by three exercise sessions using the NeuroMotion system, with post-session questionnaires and structured interviews.

The System Usability Scale (SUS) score averaged 82.4 out of 100, indicating excellent usability. Participants particularly appreciated the real-time feedback and the intuitive visual presentation of posture status. The multilingual voice feedback received positive feedback from non-native English speakers.

Post-testing interviews revealed that 92% of participants would use the system regularly if available. Suggestions for improvement focused on expanding the exercise library and reducing the initial calibration time. Several participants requested additional features such as progress tracking over time and integration with wearable heart rate monitors.